



Reconstructing Humans and Objects in Interaction using Large Reconstruction Models

Agniv Chatterjee Georgios Pavlakos
The University of Texas at Austin



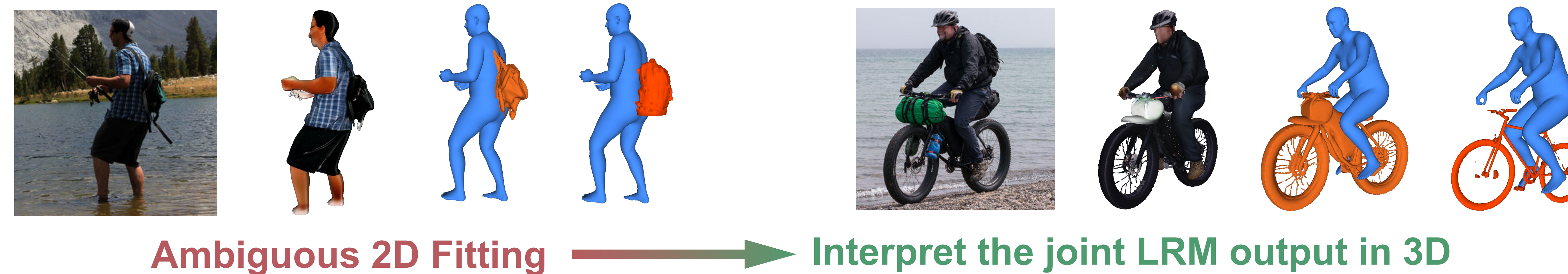
Objective

Given a single RGB image, the goal is to recover the 3D human mesh, the object geometry, and their relative 3D arrangement without **contact labels** or **object templates**

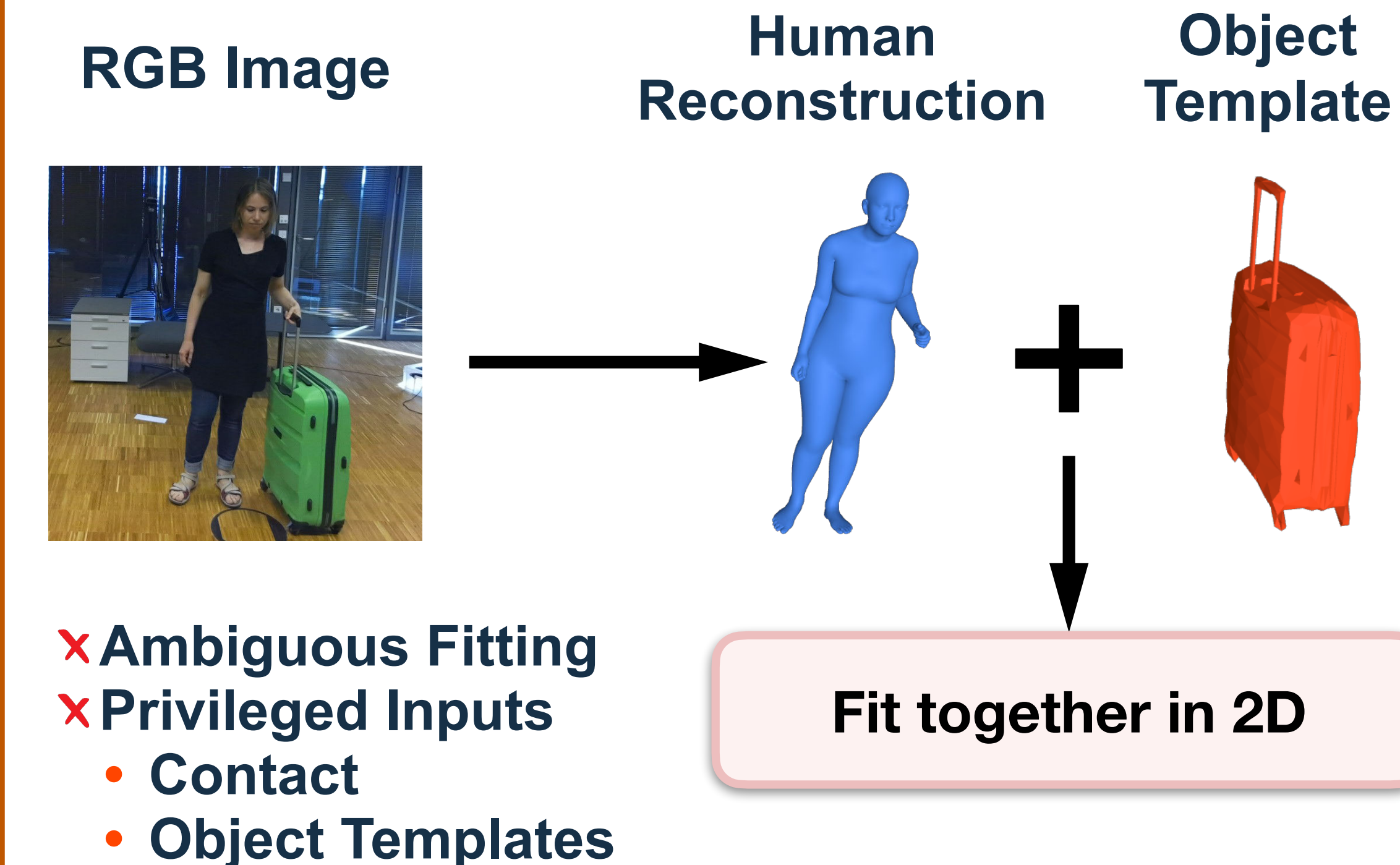


Key Insight

A joint reconstruction preserves the interaction before parametric fitting

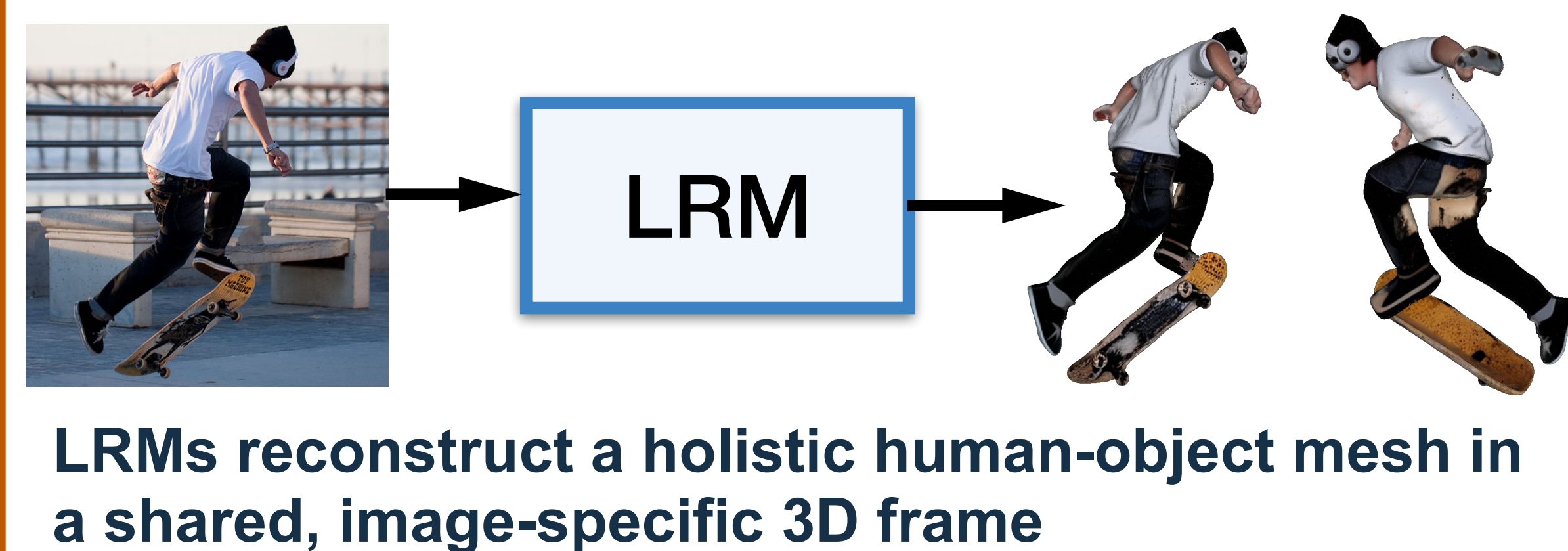


Existing Methods



Can we recover the interaction in 3D first?

Large Reconstruction Models



Why the LRM output is useful?

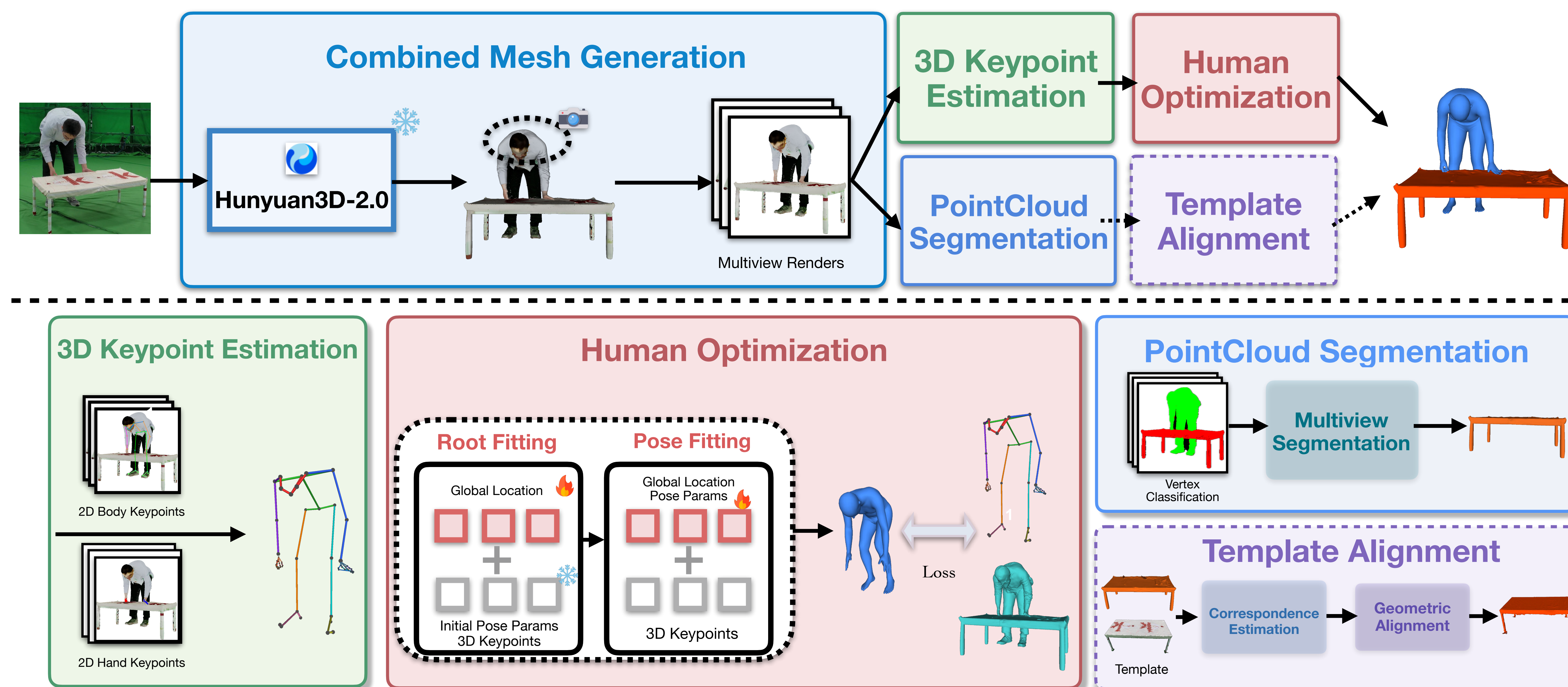
A single image-specific 3D scaffold

Relative Arrangement Proximity Shared 3D coordinate frame

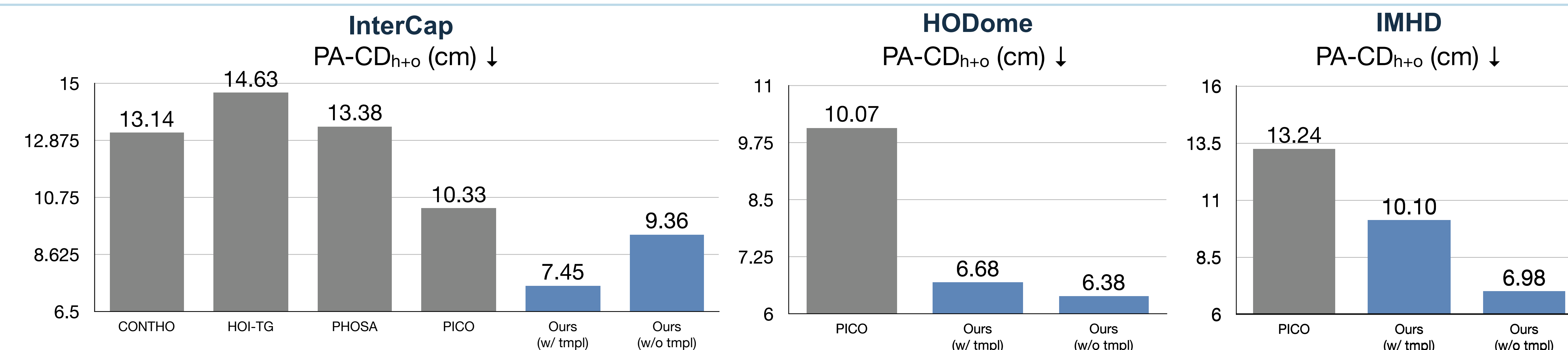
Fit separate models to 2D vs Explain a joint reconstruction in 3D

MILO: Method Overview

The LRM remains fixed; MILO explains its geometry using **multiview evidence**, **parametric priors**, and **optional object templates**.

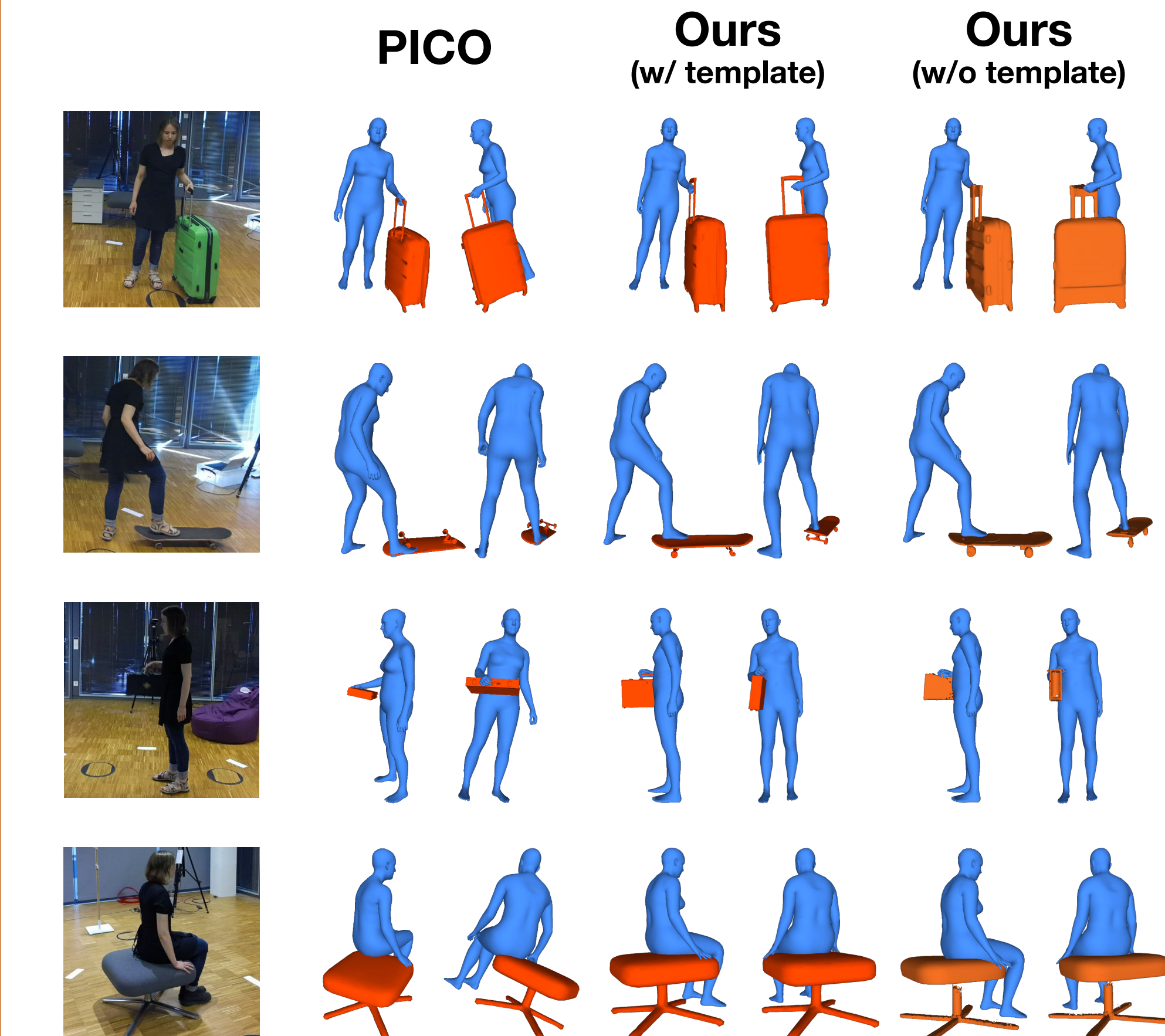


Quantitative Results

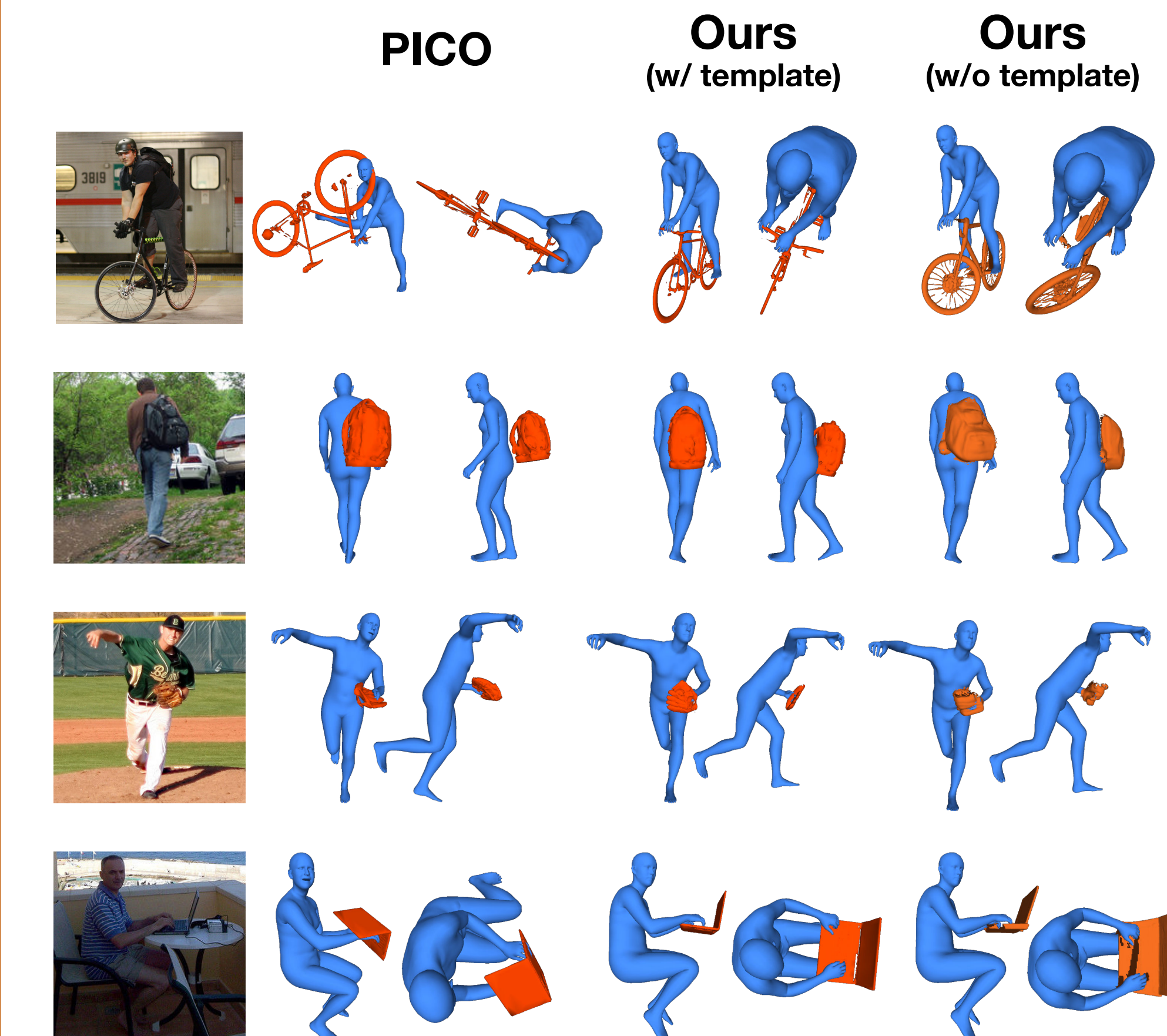


Qualitative Results

InterCap



PICO-db: in-the-wild



Limitations

Performance bounded by LRM reconstruction quality
Point-cloud segmentation can fail due to noisy masks

